ALGEBRAIC NUMBER THEORY

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Contents

1. Dedekind domains

Definition 1.0.1 (Noetherian conditions for a ring)**.** Let R be a ring. We say:

- R is *left-Noetherian* if every ascending chain of left ideals of R stabilizes, i.e., if for any chain
$$I_1 \subseteq I_2 \subseteq I_3 \subseteq \cdots$$
of left ideals, there exists n such that $I_m = I_n$ for all $m \geq n$. That is, R satisfies the ACC on the left ideals.
- R is *right-Noetherian* if every ascending chain of right ideals of R stabilizes. That is, R satisfies the ACC on the right ideals.
- R is *Noetherian* if it is both left-Noetherian and right-Noetherian.

If R is commutative (Definition C.1.1), then R is Noetherian if and only if every ideal is finitely generated.

Definition 1.0.2 (Discrete valuation ring)**.** A local (Definition C.1.5) integral domain $(R, \mathfrak{m})$ with maximal ideal (Definition C.1.4) $\mathfrak{m}$ is called a *discrete valuation ring (DVR)* if $\mathfrak{m}$ is principal and nonzero, and every nonzero ideal of R is of the form $\mathfrak{m}^n$ for some integer $n \geq 0$.

A *uniformizer of R* refers to any generator of $\mathfrak{m}$.

The *(normalized) discrete valuation* $v : R^\times \to \mathbb{Z}_{\geq 1}$ is given by
$$v(x) = \text{minimal } n \text{ such that } x \in \mathfrak{m}^n.$$
Alternatively, v may be extended to a map $v : R \to \mathbb{Z}_{\geq 1} \cup \{\infty\}$ by letting $v(0) = \infty$.

(♠ TODO: make this as a proposition) In fact, v extends to a discrete valuation (Definition B.2.3) on the fraction field of R by defining
$$v\left(\frac{a}{b}\right) = v(a) - v(b)$$
for $a \in R$ and $b \in R^\times$. This is a well defined map
$$v : K \to \mathbb{Z} \cup \{\infty\}.$$
Conversely, every discrete valuation ring is the valuation ring (Definition B.2.2) of a valuation field whose valuation is discrete (Definition B.2.3).

Theorem 1.0.3 (Correspondence between DVRs and discretely valued fields)**.** Let K be a field equipped with a discrete valuation (Definition B.2.3) $v : K^\times \to \mathbb{Z}$; in particular, there is an associate nonarchimedean valuation. The valuation ring (Definition B.2.2)
$$R_v = \{x \in K \mid v(x) \geq 0\}$$
is a discrete valuation ring (Definition 1.0.2). Conversely, if $(R, \mathfrak{m})$ is a discrete valuation ring with field of fractions $K = \operatorname{Frac}(R)$, then there exists a discrete valuation
$$v : K^\times \to \mathbb{Z}$$
such that $R = R_v$ is the valuation ring associated to v.

In particular, the category of discrete valuation rings (up to isomorphism) corresponds bijectively to the category of fields equipped with a discrete valuation.

Definition 1.0.4 (Integral element over a ring)**.** Let R be a commutative ring with unity (Definition C.1.1).

1. Let A be an R-algebra. An element $a \in A$ is called *integral over R* if there exists a monic polynomial
$$p(x) = x^n + r_{n-1}x^{n-1} + \cdots + r_1 x + r_0$$
with coefficients $r_i \in R$ such that
$$p(a) = a^n + r_{n-1}a^{n-1} + \cdots + r_1 a + r_0 = 0 \quad \text{in } A.$$
2. Let A be an extension ring of R. The ring extension A/R is called an *integral extension* if every element of A is integral over R.
3. Let A be an extension ring of R. The *integral closure of R in A*, sometimes denoted by $\widetilde{A}$, is the subring
$$\widetilde{A} = \{a \in A : a \text{ is integral over } R\}.$$
We say that R is integrally closed in A if $\widetilde{A}$ coincides with A (considered as a subring of R).
4. Let R be an integral domain with field of fractions $K = \mathrm{Frac}(R)$. We say that R is *integrally closed* if it is integrally closed as a subring of K.

Definition 1.0.5 (Dedekind domain)**.** An integral domain R is called a *Dedekind domain* or a *Dedekind ring* if it satisfies the following equivalent conditions:

- R is Noetherian (Definition 1.0.1), integrally closed (Definition 1.0.4) in its field of fractions, and every nonzero prime ideal (Definition C.1.4) of R is maximal (Definition C.1.4).
- Equivalently: for every nonzero prime ideal $\mathfrak{p}$ of R, the localization (Definition C.1.3) $R_{\mathfrak{p}}$ is a discrete valuation ring (Definition 1.0.2).

The following is immediately from the definitions:

Proposition 1.0.6. Every DVR (Definition 1.0.2) is a Dedekind domain (Definition 1.0.5). Conversely, every local (Definition C.1.5) Dedekind domain is a DVR.

Proof. By definition, every DVR is a PID (Definition C.1.6). All PID's are Noetherian. All PID's are also UFD's (Definition C.1.7) which are in turn integrally closed (Definition 1.0.4) in their fields of fractions.

The ideals of a DVR $(R, \mathfrak{m})$ are of the form $\mathfrak{m}^n$ for some integer $n \geq 0$. The ideal $\mathfrak{m}^0$ is the unit ideal. The ideal $\mathfrak{m}^1$ is the maximal ideal. For $n \geq 2$, the ideal $\mathfrak{m}^n$ is not a prime ideal (Definition C.1.4) — fixing a uniformizer π of R, the element $\pi^n = \pi^{n-1} \cdot \pi$ is an element of $\mathfrak{m}^n$ and yet neither π^{n-1} nor π is an element of $\mathfrak{m}^n$.

We have thus shown that R is Noetherian, integrally closed closed whose only prime ideal is the maximal ideal, i.e. R is a Dedekind domain.

Conversely, suppose that R is a local Dedekind domain. Let $\mathfrak{m}$ be its maximal ideal.

(♠ TODO: verify the following converse argument) Since R is Noetherian and every nonzero prime ideal is maximal, R has Krull dimension at most 1. Because R is local, $\mathfrak{m}$ is the unique nonzero prime ideal. We first show that $\mathfrak{m}$ is principal. Let K be the field of fractions of R, and consider the fractional ideal

$$I = \{y \in K \mid y\mathfrak{m} \subseteq R\}.$$

Clearly $R \subseteq I$. Since $\mathfrak{m}$ contains a nonzero element and R is a domain, $I \neq (0)$. Moreover, for any $y \in I$, we have $y\mathfrak{m} \subseteq R$, which implies $y^2\mathfrak{m} = y(y\mathfrak{m}) \subseteq yR \subseteq I$ by induction; thus I is actually a subring of K. Because $\mathfrak{m}$ is finitely generated (as R is Noetherian), say $\mathfrak{m} = (a_1, \ldots, a_n)$, and because $ya_i \in R$ for all i, we have $xI \subseteq R$ for any nonzero $x \in \mathfrak{m}$ (since $xy = yx \in y\mathfrak{m} \subseteq R$). Hence $I \subseteq x^{-1}R$. As R is Noetherian, the submodule I of the free module $x^{-1}R \cong R$ is finitely generated. For any $y \in I$, multiplication by y maps the finitely generated faithful R-module I into itself, so by the determinant trick (or Cayley-Hamilton), y satisfies a monic polynomial over R. Thus every element of I is integral over R. Since R is integrally closed in K, we conclude $I = R$. This means $\mathfrak{m}^{-1}\mathfrak{m} = R$, so $\mathfrak{m}$ is an invertible ideal.

In a local ring, every invertible ideal is principal (this follows immediately from Nakayama's Lemma: if $\mathfrak{m}/\mathfrak{m}^2$ required more than one generator over $R/\mathfrak{m}$, invertibility would fail). Hence $\mathfrak{m} = (\pi)$ for some uniformizer π.

Now let $J \neq (0)$ be any nonzero ideal of R. Since $\dim R \leq 1$, the only prime ideals are (0) and $\mathfrak{m}$, so $\sqrt{(x)} = \mathfrak{m}$ for every $0 \neq x \in J$. Thus there exists a largest integer $n \geq 0$ such that $J \subseteq \mathfrak{m}^n = (\pi^n)$. Writing $J = \pi^n K$ for some ideal K, we have $K \not\subseteq \mathfrak{m}$, which forces $K = R$ (as $\mathfrak{m}$ is the unique maximal ideal). Therefore $J = (\pi^n)$, proving that every nonzero ideal of R is principal. Consequently, R is a local PID. A local PID with Krull dimension 1 is precisely a discrete valuation ring by definition. This completes the proof.

□

Proposition 1.0.7. Let R be a Dedekind domain (Definition 1.0.5). It is a PID (Definition C.1.6) if and only if it is a UFD (Definition C.1.7).

2. Local and global fields

2.1. **Local fields.**

Definition 2.1.1 (Local field)**.** A *local field* is a field K with a nontrivial absolute value (Definition B.1.1) $|\cdot|$ such that K is locally compact (Definition C.2.1) under the topology induced by $|\cdot|$ (Definition C.2.4); we almost always treat a local field as a topological field with this topology.

The local field K is called *archimedean* if its absolute value is archimedean (Definition B.1.5) and is called *non-archimedean* if its absolute value is non-archimedean (Definition B.1.5).

(♠ TODO: p-adic numbers) (♠ TODO: laurent series) It is worth noting (Theorem 2.1.4) that archimedean local fields are isomorphic to either $\mathbb{R}$ or $\mathbb{C}$ whereas non-archimedean local fields are isomorphic to finite extensions of $\mathbb{Q}_p$ or $\mathbb{F}_q((t))$ for a prime p or a prime power q.

Convention 2.1.2. Given a local field (Definition 2.1.1) K with absolute value $|\cdot|$, we almost always equip it with the metric induced by $|\cdot|$ (Definition B.1.4), which is a complete metric (Theorem 2.1.3)

Theorem 2.1.3. Let K be a local field (Definition 2.1.1). K is complete under the metric induced by (Definition B.1.4) its absolute value.

Theorem 2.1.4. (♠ TODO: TODO: define Laurent seires) Up to isomorphism, local fields (Definition 2.1.1) consists precisely of:

- $\mathbb{R}$ and $\mathbb{C}$, which are the archimedean (Definition B.1.5) local fields),
- finite extensions of $\mathbb{Q}_p$ for a prime p, which are the nonarchimedean local fields of characteristic 0, and
- finite extensions of $\mathbb{F}_p((t))$, the field of formal Laurent series over a finite field, which are the non-archimedean local fields of positive characteristic.

2.2. Global fields.

Definition 2.2.1. A *number field* is a finite extension of the field of rational numbers $\mathbb{Q}$.

Definition 2.2.2. A *global function field* is a finite extension of the field of rational functions (Definition C.1.8) $\mathbb{F}_q(t)$ in one variable over a finite field $\mathbb{F}_q$.

Definition 2.2.3. A *global field* is a field K that is either:

- a number field (Definition 2.2.1), i.e. a finite extension of $\mathbb{Q}$, or
- a global function field (Definition 2.2.2), i.e. a finite extension of the field of rational functions (Definition C.1.8) $\mathbb{F}_q(t)$ in one variable over a finite field $\mathbb{F}_q$.

Definition 2.2.4. Two absolute values (Definition B.1.1) $|\cdot|_1$ and $|\cdot|_2$ on a field F are *equivalent* if there exists a positive real number $c > 0$ such that

$$|\cdot|_1 = |\cdot|_2^c.$$

Definition 2.2.5. Let K be a field. A *place of K* is some function of the form $P : K \to L \cup \{\infty\}$ where L is a field satisfying:

1. $P(0) = 0$
2. $P(1) = 1$.
3. $P(x + y) = P(x) + P(y)$ whenever $P(x) + P(y)$ is defined.
4. $P(x \cdot y) = P(x) \cdot P(y)$ whenever $P(x) \cdot P(y)$ is defined.

where ∞ satisfies the following rules:

1. $x + \infty = \infty$ for all $x \in L$
2. $x \cdot \infty = \infty$ for $x \neq 0$.
3. $\infty + \infty$ is undefined

4. $0 \cdot \infty$ is undefined.

It should be noted that there are conventions that define $\infty + \infty$ as ∞.

The above notion of place of a field should not be mistaken for the notion of a place of a global field (Definition 2.2.7), which are described by absolute values

Definition 2.2.6. Let K be a field. Two places $P_1 : K \to L_1 \cup \{\infty\}$ and $P_2 : K \to L_2 \cup \{\infty\}$ are said to be equivalent if there exists a field isomorphism $\sigma : \mathrm{im}(P_1) \cap L_1 \to \mathrm{im}(P_2) \cap L_2$ such that $P_2 = \sigma \circ P_1$, where σ is extended to $\mathrm{im}(P_1) \cup \{\infty\}$ by $\sigma(\infty) = \infty$.

Definition 2.2.7 (Place of a global field)**.** Let F be a global field (Definition 2.2.3). A *place of F* is an equivalence (Definition 2.2.4) class (Definition 2.2.4) of absolute values (Definition B.1.1) on F.

If any (equivalently all) representatives of a place v of F is an archimedean absolute value (Definition B.1.5) (resp. non-archimedean absolute value), then we say that v is an *archimedean place* (resp. *non-archimedean place*). A representative of a place v is often denoted by $|\cdot|_v$.

In the case that F is a number field (Definition 2.2.1), and archimedean place is called an *infinite place* and a non-archimedean place is called a *finite place*. This number field terminology should not be confused with terminology for global function fields (Definition 2.2.2) — in the case that F is a global function field, all places of F are non-archimedean and yet one may refer to some (finitely many) places of F as *"infinite"* with respect to “presentation/choice of a transcendental coordinate function” of F.

The above notion of place of a global field should not be mistaken for the notion of a place of a general field (Definition 2.2.5), which are not described by absolute values.

Definition 2.2.8. Let K be a global field (Definition 2.2.3) and let v be a place (Definition 2.2.7) of K. Write $|\cdot|_v$ for an absolute value (Definition B.1.1) representing v.

The *completion of K at v*, often denoted K_v, is the completion of K with respect to the metric induced by $|\cdot|_v$ (Definition B.1.4).

2.3. **Rings of integers of local and global fields.**

Definition 2.3.1. Let R be a Dedekind domain (Definition 1.0.5) with fraction field K. Let L be a field extension of K. The *ring of integers of L over R* is the integral closure (Definition 1.0.4), denoted $\mathcal{O}_L$, of R in L.

If L/K is a finite algebraic extension, then $\mathcal{O}_L$ is itself a Dedekind domain.

Definition 2.3.2 (Ring of Integers)**.** Let K be either a global field (Definition 2.2.3) or a nonarchimedean (Definition B.1.5) local field (Definition 2.1.1).

The *ring of integers of K* or *integer ring of K*, often denoted by notations including $\mathcal{O}_K$ and $\mathscr{O}_K$, is the set of elements of K that are integral over an appropriate base ring as follows:

- In the case where K is a number field, $\mathcal{O}_K$ is the integral closure (Definition 1.0.4) of $\mathbb{Z}$ in K, i.e.

$$\mathcal{O}_K := \{\, x \in K \mid x \text{ is a root of a monic polynomial } f(t) \in \mathbb{Z}[t] \,\}.$$

  Equivalently, $\mathcal{O}_K$ is the ring of integers of K over $\mathbb{Z}$ (Definition 2.3.1).
- In the case where K is a global function field (Definition 2.2.2) over a finite field, i.e. a finite extension of $\mathbb{F}_q(t)$ (Definition C.1.8), $\mathcal{O}_K$ is the integral closure of the polynomial ring $\mathbb{F}_q[t]$ in K.

  Equivalently, $\mathcal{O}_K$ is the ring of integers of K over the polynomial ring $\mathbb{F}_q[t]$ (Definition 2.3.1).
- In the case where K is a nonarchimedean (Definition B.1.5) local field (Definition 2.1.1) with discrete valuation (Definition B.2.3) v, $\mathcal{O}_K$ is the valuation ring $\mathcal{O}_v$ (Definition B.2.2) of v, i.e.

$$\mathcal{O}_K := \mathcal{O}_v = \{x \in K \mid v(x) \geq 0\}.$$

  Equivalently, identifying K with (Theorem 2.1.4) a finite extension of $\mathbb{Q}_p$ or $\mathbb{F}_p((t))$, the ring $\mathcal{O}_K$ is the integral closure of $\mathbb{Z}_p$ or of $\mathbb{F}_p[[t]]$ in K. (♠ TODO:) In other words, $\mathcal{O}_K$ is the ring of integers of K over $\mathbb{Z}_p$ or over $\mathbb{F}_p[[t]]$ (Definition 2.3.1).

Theorem 2.3.3.
1. The ring of integers (Definition 2.3.2) of a nonarchimedean local field (Definition 2.1.1) is a DVR (Definition 1.0.2). In particular (Proposition 1.0.6), it is a Dedekind domains (Definition 1.0.5).
2. The ring of integers (Definition 2.3.2) of a global field (Definition 2.2.3)is a Dedekind domains (Definition 1.0.5).

2.4. **Fractional ideals and the ideal class group.**

Definition 2.4.1 (Fractional Ideal of an Integral Domain)**.** Let R be an integral domain and let $K = \mathrm{Frac}(R)$ be its field of fractions.

A *fractional ideal of R* is a nonzero R-submodule $I \subseteq K$ such that there exists a nonzero element $d \in R$ with the property that $dI \subseteq R$. In other words, I can be cleared of denominators by multiplying by some nonzero element of R.

The set of all fractional ideals of R is denoted by $\mathrm{FracId}(R)$. Under multiplication defined by

$$IJ = \left\{ \sum_{k=1}^{n} x_k y_k \,\middle|\, x_k \in I,\ y_k \in J,\ n \geq 0 \right\},$$

the set $\mathrm{FracId}(R)$ forms a commutative monoid. The identity element is R itself, viewed as a fractional ideal.

A fractional ideal I is said to be *invertible* if there exists a fractional ideal J such that $IJ = R$. The set of all invertible fractional ideals is denoted by $\mathrm{InvFracId}(R)$.

Definition 2.4.2 (Principal Fractional Ideal)**.** Let R be an integral domain and let $K = \mathrm{Frac}(R)$ be its field of fractions. Let $\mathrm{FracId}(R)$ denote the set of fractional ideals (Definition 2.4.1) of R.

A fractional ideal I of R is called a *principal fractional ideal* if there exists an element $\alpha \in K^\times$ such that $I = (\alpha) = \alpha R$. The set of all principal fractional ideals is denoted by $\mathrm{PFracId}(R)$.

The map $K^\times \to \mathrm{FracId}(R)$ given by $\alpha \mapsto (\alpha)$ is a surjective homomorphism from the multiplicative group $K^\times$ onto $\mathrm{PFracId}(R)$, with kernel $R^\times$. Thus $\mathrm{PFracId}(R) \cong K^\times/R^\times$ as groups.

Definition 2.4.3 (Ideal Class Group of a Dedekind Domain)**.** Let R be a Dedekind domain (Definition 1.0.5) with field of fractions K. Let $\mathrm{FracId}(R)$ denote the monoid of fractional ideals (Definition 2.4.1) of R, let $\mathrm{InvFracId}(R) \subseteq \mathrm{FracId}(R)$ denote the subgroup of invertible fractional ideals (Definition 2.4.1), and let $\mathrm{PFracId}(R) \subseteq \mathrm{InvFracId}(R)$ denote the subgroup of principal fractional ideals (Definition 2.4.2).

The *ideal class group of R*, denoted by $\mathrm{Cl}(R)$, is defined as the quotient group

$$\mathrm{Cl}(R) = \mathrm{InvFracId}(R)/\,\mathrm{PFracId}(R).$$

The order of $\mathrm{Cl}(R)$, denoted by h_R or $h(R)$, is called the *class number of R*. The class group measures the failure of unique factorization in R: R has unique factorization (Definition C.1.7) into irreducible elements (i.e., R is a principal ideal domain (Definition C.1.6)) if and only if $\mathrm{Cl}(R)$ is trivial.

When $R = \mathcal{O}_K$ is the ring of integers (Definition 2.3.2) of a number field (Definition 2.2.1) K, we write $\mathrm{Cl}(K)$ for $\mathrm{Cl}(\mathcal{O}_K)$.

2.5. **Discriminants.**

Definition 2.5.1 (Discriminant of a Number Field)**.** Let K be a number field (Definition 2.2.1) of degree $n = [K : \mathbb{Q}]$, and let $\sigma_1, \dots, \sigma_n : K \hookrightarrow \mathbb{C}$ be the distinct embeddings of K into $\mathbb{C}$.

1. For any basis $\{\alpha_1, \dots, \alpha_n\}$ of K as a vector space over $\mathbb{Q}$, the discriminant of this basis is defined by
$$\mathrm{disc}(\alpha_1, \dots, \alpha_n) = \det\bigl(\sigma_i(\alpha_j)\bigr)^2_{1\le i,j\le n}.$$
2. The *discriminant of K*, denoted by d_K or $\mathrm{disc}(K)$, is the discriminant of any $\mathbb{Q}$-basis of the ring of integers $\mathcal{O}_K$. Equivalently, if $\{\omega_1, \dots, \omega_n\}$ is an integral basis for $\mathcal{O}_K$ over $\mathbb{Z}$, then
$$d_K = \det\bigl(\sigma_i(\omega_j)\bigr)^2_{1\le i,j\le n}.$$
The discriminant d_K is a nonzero integer and is independent of the choice of integral basis. It satisfies $d_K \equiv 0, 1 \pmod 4$.

2.6. **Zeta functions.**

Definition 2.6.1. Let $s \in \mathbb{C}$. The *Riemann zeta function* $\zeta(s)$ is defined as

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s},$$

which converges absolutely when $\Re(s) > 1$ (Lemma 2.6.2), and in this case $\zeta(s)$ can be expressed as an Euler product:

$$\zeta(s) = \prod_{n=1}^{\infty} \frac{1}{(1 - p^{-s})}.$$

(♠ TODO: TODO: discuss that it extends meromorphically, its trivial zeroes, functional equation)

Here are alternate definitions of $\zeta(s)$:

1. It is a special case of the Dirichlet L-function $L(s, \chi)$ in case that $\chi : \mathbb{Z} \to \mathbb{C}$ is the trivial Dirichlet character (modulo 0) such that $\chi(n) = 1$ for all integers $n \neq 0$.
2. It is a special case of the Dedekind zeta function $\zeta_K(s)$ (Definition 2.6.3) in the case that the number field K is simply the field $\mathbb{Q}$ of rational numbers (Definition 2.6.3).

Lemma 2.6.2. (♠ TODO: AI generated, verify) Let $s \in \mathbb{C}$ with $\Re(s) > 1$. The Riemann zeta function (Definition 2.6.1)

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$$

converges absolutely on this region.

Proof. (♠ TODO: AI generated, verify) Write $s = \sigma + it$ with $\sigma, t \in \mathbb{R}$ and $\sigma > 1$. The complex absolute value of n^{-s} is

$$|n^{-s}| = |e^{-s \log n}| = |e^{-(\sigma+it) \log n}| = e^{-\sigma \log n} |e^{-it \log n}| = n^{-\sigma},$$

since $|e^{-it \log n}| = 1$. Thus the series of norms is $\sum_{n=1}^{\infty} n^{-\sigma}$.

We show that $\sum_{n=1}^{\infty} n^{-\sigma}$ converges for $\sigma > 1$. For $\sigma > 0$, the function $x \mapsto x^{-\sigma}$ is decreasing on $[1, \infty)$, so by the integral bound

$$\sum_{n=2}^{\infty} n^{-\sigma} \leq \int_1^{\infty} x^{-\sigma}\, dx = \left[\frac{x^{1-\sigma}}{1-\sigma} \right]_1^{\infty} = \frac{1}{\sigma - 1}.$$

Since $\sigma > 1$, the improper integral converges, and therefore the series $\sum_{n=2}^{\infty} n^{-\sigma}$ is bounded above. Adding the $n = 1$ term $1^{-\sigma} = 1$, the full series $\sum_{n=1}^{\infty} n^{-\sigma}$ also converges.

Hence $\sum_{n=1}^{\infty} |n^{-s}|$ converges, i.e., the Riemann zeta function converges absolutely for $\Re(s) > 1$. □

Definition 2.6.3. Let K be a number field (Definition 2.2.1), and let $s \in \mathbb{C}$ with $\Re(s) > 1$. The *Dedekind zeta function of K* is defined as

$$\zeta_K(s) = \sum_{\mathfrak{a} \neq (0)} \frac{1}{N_{K/\mathbb{Q}}(\mathfrak{a})^s},$$

where $\mathfrak{a}$ runs over all nonzero ideals of $\mathcal{O}_K$. (♠ TODO: TODO: discuss the Euler product) (♠ TODO: TODO: discuss that it extends meromorphically, its trivial zeroes, functional equation)

2.7. **Local fields are exactly the completions of global fields at places.**

Theorem 2.7.1. Let L be a global field (Definition 2.2.3) and let v be any place (Definition 2.2.7). The completion L_v (Definition 2.2.8) of L with respect to v is a local field (Definition 2.1.1).

Remark 2.7.2. Conversely, all local fields can be obtained as completions of global fields at places.

Definition 2.7.3. Let K be a global field (Definition 2.2.3) or a local field (Definition 2.1.1), let $\mathcal{O}_K$ be its ring of integers (Definition 2.3.2), and let $I \subseteq \mathcal{O}_K$ be a nonzero ideal. The *(absolute) norm of I* is defined as the cardinality of R/I, which is always finite.

Definition 2.7.4. Let K be a number field with ring of integers (Definition 2.3.2) $\mathcal{O}_K$. Let $\mathfrak{p}$ be a nonzero prime ideal (Definition C.1.4) of $\mathcal{O}_K$.

1. The *(normalized)* $\mathfrak{p}$*-adic valuation on* K refers to the discrete valuation (Definition B.2.3) $v_\mathfrak{p}$ defined by
$$v_\mathfrak{p}(a) = \max\{n : a \in \mathfrak{p}^n\}$$
for $a \in \mathcal{O}_K \setminus \{0\}$ and
$$v_\mathfrak{p}(x) = v_\mathfrak{p}(a) - v_\mathfrak{p}(b)$$
if $x = \frac{a}{b}$ for $a, b \in \mathcal{O}_K \setminus \{0\}$. This is well defined.
2. The $\mathfrak{p}$*-adic absolute value on* K refers to the non-archimedean (Definition B.1.5) absolute value (Definition B.1.1) given by
$$|x|_\mathfrak{p} = N(\mathfrak{p})^{-v_\mathfrak{p}(x)}$$
(Definition 2.7.3).

Theorem 2.7.5.
1. Let K be a number field (Definition 2.2.1) with ring of integers (Definition 2.3.2) $\mathcal{O}_K$.
   (a) The nonarchimedean (Definition B.1.5) absolute values (Definition B.1.1) on K are all equivalent to (Definition 2.2.4) $\mathfrak{p}$-adic absolute values (Definition 2.7.4) for nonzero prime ideals (Definition C.1.4) $\mathfrak{p}$ of $\mathcal{O}_K$.
   (b) The archimedean absolute values on K are equivalent to absolute values induced by an embedding $\sigma : K \hookrightarrow \mathbb{C}$ (among these may be embeddings whose images lie in $\mathbb{R}$).
2. Let K be a global function field (Definition 2.2.2) with ring of integers (Definition 2.3.2) $\mathcal{O}_K$.
   All absolute values (Definition B.1.1) on K are non-archimedean (Definition B.1.5).

2.8. **Ramification and inertia of extensions of local and global fields.**

Definition 2.8.1. Let L/K be an extension of valued fields (Definition B.2.1) with valuations $v : K \to \Gamma_K \cup \{\infty\}$ and $w : L \to \Gamma_L \cup \{\infty\}$. We say that w is an *extension of* v if

1. Γ_K is an ordered subgroup of Γ_L
2. For all $x \in K$, $w(x) = v(x)$.

In other words, $v = w|_K$.

It is appropriate to refer to $(L, w)/(K, v)$ as an *extension of valuation fields*.

Theorem 2.8.2 (Chevalley's extension theorem)**.** Let L/K be an extension of fields. Let v be a valuation (Definition B.2.1) on K. There exists at least one valuation of L that extends (Definition 2.8.1) v.

Definition 2.8.3. Let K be a field. Let v be a non-archimedean valuation (Definition B.1.5) on K with residue field (Definition B.2.2) κ. The field K is said to be *henselian with respect to a valuation v* if it satisfies any of the following equivalent conditions:

1. For every algebraic extension L/K, there exists a unique extension (Definition 2.8.1) of v to L.
2. For every polynomial $f \in \mathcal{O}_K[x]$ and every root $\alpha_0 \in \kappa$ of the reduction $\bar{f} \in \kappa[x]$ such that $\bar{f}'(\alpha_0) \neq 0$, there exists a unique root $\alpha \in \mathcal{O}_K$ of f such that $\bar{\alpha} = \alpha_0$.
3. For every $f \in \mathcal{O}_K[x]$, if $\bar{f} = g_0 h_0$ for some coprime polynomials $g_0, h_0 \in \kappa[x]$, then there exist $g, h \in \mathcal{O}_K[x]$ such that $f = gh$ with $\bar{g} = g_0$, $\bar{h} = h_0$, and $\deg(g) = \deg(g_0)$.
4. Every finite K-algebra is a product of Henselian local rings.

Proposition 2.8.4. Let (K, v) be a valuation field (Definition B.2.1). It is Henselian (Definition 2.8.3) if and only if its valuation ring (Definition B.2.2) is a Henselian (local) ring.

Context 2.8.5. Let L/K be an extension of fields, let $w : L \to \Gamma_L \cup \{\infty\}$, $v : K \to \Gamma_K \cup \{\infty\}$ be valuations (Definition B.2.1) on L and K with w extending (Definition 2.8.1) v and w and v assumed to be surjective. Let $\mathcal{O}_w$ and $\mathcal{O}_v$ be their respective valuation rings (Definition B.2.2), which are local (Definition C.1.5), with maximal ideals (Definition C.1.4) $\mathfrak{m}_w$ and $\mathfrak{m}_v$, and let k_w and k_v be their residue fields.

Definition 2.8.6. Assume Context 2.8.5. The *ramification index of w over v*, denoted $e(w/v)$, is the index of the value group Γ_v in Γ_w:

$$e(w/v) = [\Gamma_w : \Gamma_v]$$

Definition 2.8.7. Assume Context 2.8.5. The *inertial degree index of w over v*, denoted $f(w/v)$, is the degree of the residue field extension:

$$f(w/v) = [k_w : k_v]$$

Definition 2.8.8. Assume Context 2.8.5.

1. Further assume that L/K is finite. The extension $(L, w)/(K, v)$ of valuation fields is *unramified* if
   (a) $e(w/v) = 1$ (Definition 2.8.6)
   (b) $f(w/v) = [L : K]$ (Definition 2.8.7)
   (c) The residue field extension k_w/k_v is separable.
2. Removing the assumption that L/K is finite, the extension $(L, w)/(K, v)$ of valuation fields is *unramified* if for every finite subextension $K \subseteq E \subseteq L$, $(E, w|_E)/(K, v)$ is unramified in the above sense. Equivalently, $(L, w)/(K, v)$ is unramified if the following hold:
   (a) The natural inclusion of value groups is an isomorphism: $\Gamma_w = \Gamma_v$.

(b) The ideal generated by $\mathfrak{m}_v$ in $\mathcal{O}_w$ is the maximal ideal of $\mathcal{O}_w$: $\mathfrak{m}_v\mathcal{O}_w = \mathfrak{m}_w$.
(c) The residue field extension k_L/k_K is a separable algebraic extension.

In the case that the valuation is of rank 1 (Definition B.2.1), the valuation field extension is unramified if and only if the extension of the associated (Definition B.2.4) absolute valued field is unramified.

Definition 2.8.9. Assume Context 2.8.5.

1. Further assume that L/K is finite. The extension $(L, w)/(K, v)$ of valuation fields is *totally ramified* if
   (a) $e(w/v) = [L : K]$ (Definition 2.8.6)
   (b) $f(w/v) = 1$ (Definition 2.8.7)
2. Removing the assumption that L/K is finite, the extension $(L, w)/(K, v)$ of valuation fields is *totally ramified* if for every finite subextension $K \subseteq E \subseteq L$, $(E, w|_E)/(K, v)$ is totally ramified in the above sense.

Definition 2.8.10. Assume Context 2.8.5 and let p be the characteristic of the residue field k_v.

1. Further assume that L/K is finite.
   (a) The extension $(L, w)/(K, v)$ is *tamely ramified* if:
      (i) The residue field extension k_w/k_v is separable;
      (ii) If $p > 0$, then $p \nmid e(w/v)$;
      (iii) The extension is defectless, i.e., $[L : K] = e(w/v)f(w/v)$.
   (b) The extension $(L, w)/(K, v)$ is *wildly ramified* if it is not tamely ramified.
      - *Note:* For non-archimedean local fields (Definition 2.1.1), this occurs if and only if $p \mid e(w/v)$.
2. Removing the assumption that L/K is finite, the extension $(L, w)/(K, v)$ is *tamely ramified* (resp. *wildly ramified*) if for every (resp. for some) finite subextension $K \subseteq E \subseteq L$, $(E, w|_E)/(K, v)$ is tamely ramified (resp. wildly ramified).

2.8.1. *Valuations at points of schemes.*

Definition 2.8.11. Let K be a field. Let $(A, \mathfrak{m}_A)$ and $(B, \mathfrak{m}_B)$ be local rings (Definition C.1.5) contained in K. We say that B *dominates A* if

- $A \subseteq B$ and
- $\mathfrak{m}_B \cap A = \mathfrak{m}_A$.

Definition 2.8.12.
1. Let R be an integral domain. Let $\mathfrak{p}$ be a prime ideal of R (Definition C.1.4). Let v be a non-archimedean valuation (Definition B.1.5) on the fraction field of R.

   We say that v is *centered at $\mathfrak{p}$* if the valuation ring (Definition B.2.2) $\mathcal{O}_v$ dominates (Definition 2.8.11) $R_{\mathfrak{p}}$ (Definition C.1.3).
2. Let X be an integral scheme. Let $x \in X$ be a point. Let v be a valuation (Definition B.2.1) on the function field of X. We say that v is *centered at x* if the valuation ring (Definition B.2.2) $\mathcal{O}_v$ dominates (Definition 2.8.11) $\mathcal{O}_{X,x}$.

Note that for an integral domain R, prime ideal $\mathfrak{p}$, and a valuation v of $\mathrm{Frac}(R)$, the valuation v is centered at $\mathfrak{p}$ if and only if it is centered at the point of $\operatorname{Spec} R$ corresponding to $\mathfrak{p}$.

Proposition 2.8.13. Let R be an integral domain. for any prime ideal (Definition C.1.4) $\mathfrak{p}$, there is at least one valuation v centered at (Definition 2.8.12) $\mathfrak{p}$.

Proposition 2.8.14. Let X be an integral scheme. For any $x \in X$, there is at least one valuation (Definition B.2.1) v centered at (Definition 2.8.12) x.

Theorem 2.8.15. Let X be an integral scheme. If x is a normal pointof X of codimension 1, then there is a unique discrete valuation (Definition B.2.3) v centered at (Definition 2.8.12) x.

2.8.2. *Decomposition groups and inertia groups.*

Definition 2.8.16. Assume Context 2.8.5 and further assume that L/K is a finite Galois extension (Definition A.0.1).

The *decomposition group of L/K* is the subgroup

$$D_w = \{\sigma \in \mathrm{Gal}(L/K) : w \circ \sigma = w\}$$

(Definition A.0.1)

Definition 2.8.17. Assume Context 2.8.5 and further assume that L/K is a finite Galois extension (Definition A.0.1).

There is a natural surjective homomorphism

$$D_w \to \mathrm{Gal}(k_w/k_v)$$

on the residue fields. The *inertia group of L/K* is its kernel:

$$I_w = \{\sigma \in D_w : \sigma(x) \equiv x \pmod{\mathfrak{m}_w} \text{ for all } x \in \mathcal{O}_w\}$$

(Definition 2.8.16)

Definition 2.8.18. Assume Context 2.8.5 and further assume that L/K is a Galois extension (Definition A.0.1).

The *decomposition group of L/K* is the subgroup

$$D_w = \{\sigma \in \mathrm{Gal}(L/K) : w \circ \sigma = w\}.$$

(Definition A.0.1) It is also the inverse limit of the decomposition groups (Definition 2.8.16) of the finite Galois subextensions E/K within L (with valuation restricted from the valuation of L). It is a closed subgroup of $\mathrm{Gal}(L/K)$.

Letting (K, v) be a valuation field, its *decomposition group* $D_{K,v}$ (also sometimes denoted by $G_{K,v}$) is the decomposition group of K^{sep}/K in the above sense where a valuation on K^{sep} extending v is fixed.

Definition 2.8.19. Assume Context 2.8.5 and further assume that L/K is a Galois extension (Definition A.0.1).

There is a natural surjective homomorphism

$$D_w \to \mathrm{Gal}(k_w/k_v)$$

(Definition 2.8.18) on the residue fields. The *inertia group of L/K* is its kernel:

$$I_w = \{\sigma \in D_w : \sigma(x) \equiv x \pmod{\mathfrak{m}_v} \text{ for all } x \in \mathcal{O}_v\}.$$

(Definition 2.8.18) (Definition A.0.1) It is also the inverse limit of the decomposition groups (Definition 2.8.16) of the finite Galois subextensions E/K within L (with valuation restricted from the valuation of L). It is a closed subgroup of D_w.

Letting (K, v) be a valuation field, its *inertia group* $I_{K,v}$ is the inertia group of K^{sep}/K in the above sense where a valuation on K^{sep} extending v is fixed.

Definition 2.8.20. Assume Context 2.8.5 and further assume that L/K is a Galois extension (Definition A.0.1). Assume moreover that the residue field extension k_w/k_v is finite (equivalently, the extension L/K is unramified (Definition 2.8.8) at w).

Let $D_w \subset \mathrm{Gal}(L/K)$ be the decomposition group (Definition 2.8.18) and let $I_w \subset D_w$ be the inertia group (Definition 2.8.19). There is a canonical isomorphism

$$D_w/I_w \xrightarrow{\sim} \mathrm{Gal}(k_w/k_v).$$

The *arithmetic Frobenius element of L/K at w* is any element $\mathrm{Frob}_w \in D_w$ whose image in $\mathrm{Gal}(k_w/k_v)$ under this isomorphism is the *arithmetic Frobenius automorphism of k_w/k_v*, i.e., the automorphism $x \mapsto x^{|k_v|}$.

The *geometric Frobenius element of L/K at w* is the inverse $\mathrm{Frob}_w^{-1} \in D_w$; its image in $\mathrm{Gal}(k_w/k_v)$ is the *geometric Frobenius automorphism of k_w/k_v*, i.e., the automorphism $x \mapsto x^{|k_v|^{-1}}$ (equivalently, the inverse of the arithmetic Frobenius automorphism).

When L/K is finite, the Frobenius element is well-defined up to multiplication by an element of I_w; when L/K is infinite, it is well-defined up to multiplication by an element of the closure of I_w in D_w.

(Definition A.0.1) (Definition 2.8.8) (Definition 2.8.18) (Definition 2.8.19)

2.9. **Galois representations of local and global fields.**

Definition 2.9.1. Let K be a field. Let R be a topological ring.

1. Let L/K be a Galois field extension (Definition A.0.1). A *Galois representation of $\mathrm{Gal}(L/K)$ over R* refers to a continuous representation

   $$\rho : \mathrm{Gal}(L/K) \to \mathrm{Aut}_R(M)$$

   of $\mathrm{Gal}(L/K)$ as a topological group (with its profinite topology) for some topological R-module M.

   The category of Galois representations of $\mathrm{Gal}(L/K)$ over R is equivalent to the category of Galois modules of $\mathrm{Gal}(L/K)$ over R.

2. A *Galois representation of K over R* refers to a Galois representation of $\mathrm{Gal}(\overline{K}/K)$ over R.

If R is a ring and M is a module both without explicit topologies specified, then R and M are implicitly equipped with their discrete topologies.

Definition 2.9.2. Let K be a local field (Definition 2.1.1). Let R be a topological ring.

1. A *local Galois representation of K over R* is a Galois representation (Definition 2.9.1) of K over R, i.e., a continuous representation
$$\rho : G_K \longrightarrow \mathrm{Aut}_R(M)$$
of the absolute Galois group $G_K = \mathrm{Gal}(\overline{K}/K)$ (equipped with its profinite topology if K is non-archimedean) on a topological R-module M.
2. If K is an archimedean local field (Definition 2.1.1) (i.e., $K \cong \mathbb{R}$ or $K \cong \mathbb{C}$), then G_K is a finite group, so the continuity condition is automatic for any topology on M, and the representation theory of G_K reduces to the representation theory of a finite group over R.
3. If K is a non-archimedean local field, then G_K is a profinite group, and the representation is continuous if and only if the action map $G_K \times M \to M$ is continuous.

The *restriction of a local representation to the inertia subgroup* refers to the restriction of ρ to the inertia subgroup (Definition 2.8.19) $I_K \subset G_K$, and the representation is said to be *unramified* if I_K acts trivially on M (see **??**).

Definition 2.9.3. Let K be a global field (Definition 2.2.3). Let R be a topological ring.

1. A *global Galois representation of K over R* is a Galois representation (Definition 2.9.1) of K over R, i.e., a continuous representation
$$\rho : G_K \longrightarrow \mathrm{Aut}_R(M)$$
of the absolute Galois group $G_K = \mathrm{Gal}(\overline{K}/K)$ (equipped with its profinite topology) on a topological R-module M.
2. For each place v of K (Definition 2.2.7), let K_v denote the completion of K at v (Definition 2.2.8), and let $G_{K_v} = \mathrm{Gal}(\overline{K}_v/K_v)$ be the absolute Galois group of K_v. Fix an embedding $\overline{K} \hookrightarrow \overline{K}_v$ extending the natural inclusion $K \hookrightarrow K_v$; this induces a continuous injection $G_{K_v} \hookrightarrow G_K$ defined up to conjugation. The *localization of ρ at v* is the restriction
$$\rho_v : G_{K_v} \longrightarrow \mathrm{Aut}_R(M)$$
of ρ along this injection, which is a local Galois representation of K_v over R (Definition 2.9.2).
3. A global Galois representation is said to be *unramified at a place v* if its localization ρ_v at v is unramified, i.e., the inertia subgroup $I_{K_v} \subset G_{K_v}$ acts trivially on M.

The notion of a global Galois representation is the central object of study in *global Galois theory*, whose local-global compatibility and *compatible systems* (see **??**) form the foundation of the Langlands program.

2.10. **Prime decomposition in extensions of Dedekind domains.**

Definition 2.10.1. (♠ TODO: AI generated, verify) Let R be a Dedekind domain (Definition 1.0.5) with fraction field K, and let L be a finite extension of K. Let $\mathcal{O}_L$ be the ring of integers of L over R (Definition 2.3.1), which is itself a Dedekind domain. Let $\mathfrak{p}$ be a prime of R.

Since $\mathcal{O}_L$ is a Dedekind domain, the *extension of $\mathfrak{p}$ to $\mathcal{O}_L$*, i.e. the ideal $\mathfrak{p}\mathcal{O}_L$ of $\mathcal{O}_L$ generated by $\mathfrak{p}$, admits a unique factorization into prime ideals of $\mathcal{O}_L$:

$$\mathfrak{p}\mathcal{O}_L = \prod_{i=1}^{g} \mathfrak{P}_i^{e_i},$$

where the $\mathfrak{P}_i$ are distinct primes of $\mathcal{O}_L$ and each $e_i \geq 1$ is an integer. The integer e_i, also denoted $e(\mathfrak{P}_i/\mathfrak{p})$, is called the *ramification index of $\mathfrak{P}_i$ over $\mathfrak{p}$*. The inertial degree (Definition 2.8.7) f_i, also denoted $f(\mathfrak{P}_i/\mathfrak{p})$, is the degree of the residue field extension:

$$f_i = [\kappa(\mathfrak{P}_i) : \kappa(\mathfrak{p})],$$

where $\kappa(\mathfrak{p}) = R/\mathfrak{p}$ and $\kappa(\mathfrak{P}_i) = \mathcal{O}_L/\mathfrak{P}_i$ are the respective residue fields.

The *fundamental equality of prime decomposition* holds:

$$\sum_{i=1}^{g} e_i f_i = [L : K].$$

Definition 2.10.2. (♠ TODO: AI generated, verify) Assume the setting of Definition 2.10.1. Let $\mathfrak{p}$ be a prime of R with factorization $\mathfrak{p}\mathcal{O}_L = \prod_{i=1}^{g} \mathfrak{P}_i^{e_i}$ and inertial degrees $f_i = f(\mathfrak{P}_i/\mathfrak{p})$.

- $\mathfrak{p}$ is *unramified in L* if $e_i = 1$ for all i and each residue field extension $\kappa(\mathfrak{P}_i)/\kappa(\mathfrak{p})$ is separable. Otherwise $\mathfrak{p}$ is *ramified in L*.
- $\mathfrak{p}$ *splits completely in L* if $g = [L : K]$ (equivalently $e_i = f_i = 1$ for all i).
- $\mathfrak{p}$ is *inert in L* if $g = 1$ and $e_1 = 1$, so that $\mathfrak{p}\mathcal{O}_L = \mathfrak{P}$ is itself a prime of $\mathcal{O}_L$ and $f_1 = [L : K]$.
- $\mathfrak{p}$ is *totally ramified in L* if $g = 1$ and $f_1 = 1$, so that $\mathfrak{p}\mathcal{O}_L = \mathfrak{P}^{[L:K]}$.

Proposition 2.10.3. (♠ TODO: AI generated, verify) Let R be a Dedekind domain (Definition 1.0.5) with fraction field K, and let L be a finite extension of K. Let $\mathcal{O}_L$ be the ring of integers of L over R (Definition 2.3.1). Let $\mathfrak{p}$ be a prime of R with associated $\mathfrak{p}$-adic valuation $v_\mathfrak{p}$ on K.

For each prime $\mathfrak{P}$ of $\mathcal{O}_L$ appearing in the factorization $\mathfrak{p}\mathcal{O}_L = \prod \mathfrak{P}^{e(\mathfrak{P}/\mathfrak{p})}$, let $v_\mathfrak{P}$ be the $\mathfrak{P}$-adic valuation on L. Then $v_\mathfrak{P}$ is an extension of $v_\mathfrak{p}$ to L (Definition 2.8.1), and the invariants coincide:

$$e(\mathfrak{P}/\mathfrak{p}) = e(v_\mathfrak{P}/v_\mathfrak{p}), \qquad f(\mathfrak{P}/\mathfrak{p}) = f(v_\mathfrak{P}/v_\mathfrak{p}),$$

where $e(v_\mathfrak{P}/v_\mathfrak{p})$ and $f(v_\mathfrak{P}/v_\mathfrak{p})$ are the ramification index (Definition 2.8.6) and inertial degree (Definition 2.8.7) of the extension of discrete valuations.

Conversely, every extension of $v_\mathfrak{p}$ to L arises as $v_\mathfrak{P}$ for a unique prime $\mathfrak{P}$ of $\mathcal{O}_L$ lying above $\mathfrak{p}$.

3. Adèles and idèles of global fields

Definition 3.0.1. Let K be a global field (Definition 2.2.3). Write M_K for the set of all places (Definition 2.2.7) of K and write M_K^∞ for the set of archimedean places (Definition 2.2.7) of K. Let $S \subseteq M_K$ be some subset of places of K (typically, S is a finite set). For each $v \in M_K$, write $\mathcal{O}_v$ for the ring of integers (Definition 2.3.2) in the completion K_v (Definition 2.2.8) (Theorem 2.7.1)

- The *adèle ring of K*, denoted $\mathbb{A}_K$, is the restricted direct product of the K_v (over all places v of K), with respect to the $\mathcal{O}_v$ at non-archimedean (Definition B.1.5) v:

$$\mathbb{A}_K = \left\{ (x_v)_v \in \prod_{v \in M_K} K_v \;\middle|\; x_v \in \mathcal{O}_v \text{ for all but finitely many non-archimedean } v \right\}.$$

- The *idèle group of K*, commonly denoted $\mathbb{A}_K^\times$ or $\mathbb{I}_K$, is the group of invertible elements of $\mathbb{A}_K$:

$$\mathbb{I}_K = \mathbb{A}_K^\times = \left\{ (x_v)_v \in \prod_{v \in M_K} K_v^\times \;\middle|\; x_v \in \mathcal{O}_v^\times \text{ for all but finitely many non-archimedean } v \right\},$$

  where $\mathcal{O}_v^\times$ denotes the group of units of $\mathcal{O}_v$ for non-archimedean v.
- The *adèle ring outside S of K*, commonly denoted $\mathbb{A}_K^S$ or $\mathbb{A}_{K,S}$, is the restricted product of the completions K_v over all places $v \in M_K \setminus S$, with respect to the rings of integers $\mathcal{O}_v$ at non-archimedean places:

$$\mathbb{A}_{K,S} = \mathbb{A}_K^S = \left\{ (x_v)_v \in \prod_{v \in M_K \setminus S} K_v \;\middle|\; x_v \in \mathcal{O}_v \text{ for all but finitely many non-archimedean } v \right\}.$$

- The *idèle group outside S of K*, commonly denoted $(\mathbb{A}_K^\times)^S$, $(\mathbb{A}_{K,S}^\times)$, $\mathbb{I}_K^S$, or $\mathbb{I}_{K,S}$ is the group of invertible elements of $\mathbb{A}_K^S$:

$$(\mathbb{A}_K^\times)^S = \left\{ (x_v)_v \in \prod_{v \in M_K \setminus S} K_v^\times \;\middle|\; x_v \in \mathcal{O}_v^\times \text{ for all but finitely many non-archimedean } v \right\}.$$

- The *ring of finite adèles of K*, commonly denoted $\mathbb{A}_{K,\mathrm{fin}}$, $\mathbb{A}_K^{\mathrm{fin}}$, $\mathbb{A}_{K,\mathrm{f}}$, $\mathbb{A}_K^{\mathrm{f}}$, is the adèle ring outside $S = M_K^\infty$, the set of archimedean places of K:

$$\mathbb{A}_{K,\mathrm{fin}} := \mathbb{A}_K^{M_K^\infty} = \left\{ (x_v)_v \in \prod_{v \notin M_K^\infty} K_v \;\middle|\; x_v \in \mathcal{O}_v \text{ for all but finitely many non-archimedean } v \right\}.$$

- The *finite idèle group of K*, commonly denoted $\mathbb{A}_{K,\text{fin}}^\times$, $\mathbb{I}_{K,\text{fin}}$, $\mathbb{I}_K^{\text{fin}}$, $\mathbb{I}_{K,\text{f}}$, $\mathbb{I}_K^{\text{f}}$ etc. is the group of units of the ring of finite adèles:

$$\mathbb{A}_{K,\text{fin}}^\times := (\mathbb{A}_K^\times)^{M_K^\infty} = \left\{ (x_v)_v \in \prod_{v \notin M_K^\infty} K_v^\times \;\middle|\; x_v \in \mathcal{O}_v^\times \text{ for all but finitely many non-archimedean } v \right\}.$$

All of these are equipped with the restricted product topology induced by the topologies of the local fields (Definition 2.1.1) K_v and the subspace topologies thereof.

Definition 3.0.2 (Idelic norm of a global field)**.** Let F be a global field. The *idelic norm* (also called the *module*) is the map

$$|\cdot| : \mathbb{A}_F^\times \to \mathbb{R}_{>0}$$

defined by (♠ TODO: definition of canonical absolute value of places)

$$|(x_\mathfrak{p})_\mathfrak{p}| \;:=\; \prod_\mathfrak{p} |x_\mathfrak{p}|_\mathfrak{p},$$

where the product is well-defined as all but finitely many factors equal 1.

(♠ TODO: state the product formula for global fields, define idele class group) By the product formula for global fields, the idelic norm factors through the idele class group and satisfies

$$|a| = 1 \quad \text{for all } a \in F^\times,$$

where $F^\times$ is diagonally embedded in $\mathbb{A}_F^\times$.

4. Chebotarev density theorem

The Chebotarev density theorem is a statement that Frobenius elements $\mathrm{Frob}_\mathfrak{p}$ are equidistributed across the conjugacy classes of $\mathrm{Gal}(L/K)$ of a finite Galois extension of number fields, showing that the splitting behavior of primes in extensions is governed by uniform distribution with respect to the Galois group structure.

Theorem 4.0.1 (Chebotarev Density Theorem for number fields)**.** Let L/K be a finite Galois extension (Definition A.0.1) of number fields (Definition 2.2.1) with Galois group $G = \mathrm{Gal}(L/K)$ (Definition A.0.1). For a conjugacy class $C \subseteq G$, let

$$\pi_C(x) = \#\{\, \mathfrak{p} \subseteq \mathcal{O}_K : N\mathfrak{p} \le x,\ \mathfrak{p} \text{ unramified (Definition 2.10.2) in } L,\ \mathrm{Frob}_\mathfrak{p} \in C \,\}$$

be the number of prime ideals $\mathfrak{p}$ of K with norm at most x whose Frobenius conjugacy class $\mathrm{Frob}_\mathfrak{p}$ in G equals C.

(♠ TODO: define natural density of primes) Then the natural density of such primes exists and satisfies

$$\lim_{x\to\infty} \frac{\pi_C(x)}{\pi(x)} = \frac{|C|}{|G|},$$

where $\pi(x)$ is the number of prime ideals $\mathfrak{p} \subseteq \mathcal{O}_K$ with $N\mathfrak{p} \le x$.

Theorem 4.0.2 (Chebotarev Density theorem for function fields, see e.g. [Ros02, Theorem 9.13A, 9.13B])**.** Let L/K be a Galois extension (Definition A.0.1) of global function fields (Definition 2.2.2) with Galois group $G = \mathrm{Gal}(L/K)$. Let $C \subset G$ be a conjugacy class in G and S'_K be the set of primes of K which are unramified (Definition 2.10.2) in L. (♠ TODO: continue statement)

Appendix A. Galois theory

Definition A.0.1 (Galois Extension)**.** An extension L/K is called a *Galois extension* if it is both a normal extension and a separable algebraic extension. Its *Galois group*, usually denoted by $\mathrm{Gal}(L/K)$, is defined to be the automorphism group $\mathrm{Aut}(L/K)$.

Appendix B. Absolute values and valuations on fields

B.1. Absolute values on fields.

Definition B.1.1 (Absolute value on a ring)**.** Let R be a commutative ring (Definition C.1.1) and let T be an ordered semiring.

An *absolute value on R valued in T* (or a *norm on R valued in T*) is a function

$$|\cdot| : R \to T$$

that satisfies the following properties for all $a, b \in R$:

1. Positive definiteness: $|a| = 0_T$ if and only if $a = 0_R$.
2. Multiplicativity: $|ab| = |a| \cdot |b|$,
3. Subadditivity/triangle inequality: $|a + b| \leq |a| + |b|$,

A ring equipped with an absolute value may be called a *normed ring* or an *absolute valued ring*. A field equipped with an absolute value may be called a *normed field*. It may also be called a *valued field*, but this should not be confused with the more algebraic notion of a field equipped with a valuation (Definition B.2.1) with values in a more general ordered group.

In the case that T is not explicit, the absolute value $|\cdot|$ is almost always taken to be valued in $\mathbb{R}_{\geq 0}$.

Definition B.1.2 (Discrete absolute value)**.** Let K be a field equipped with an absolute value (Definition B.1.1)$|\cdot|$. The absolute value $|\cdot|$ is called *discrete* if its image $|K^\times| = \{|x| : x \in K^\times\} \subseteq \mathbb{R}_{>0}$ is a discrete subgroup of the multiplicative group $\mathbb{R}_{>0}$ (with the usual topology).

Definition B.1.3 (Trivial absolute value)**.** Let K be a field. The *trivial absolute value on K* is the absolute value (Definition B.1.1)

$$|\cdot|_{\mathrm{triv}} : K \to \mathbb{R}_{\geq 0}$$

defined by

$$|x|_{\text{triv}} := \begin{cases} 0 & \text{if } x = 0, \\ 1 & \text{if } x \neq 0. \end{cases}$$

Any absolute value on K that is not equivalent (Definition 2.2.4) to this absolute value is called *nontrivial*.

Definition B.1.4 (Metric induced by an absolute value on a commutative ring)**.** Let R be a commutative ring (Definition C.1.1) equipped with an absolute value valued in an ordered semiring (Definition B.1.1):

$$|\cdot| : R \to T.$$

The *metric induced by $|\cdot|$* is the function

$$d : R \times R \to T, \quad (a, b) \mapsto |a - b|$$

defined by

$$d(a, b) := |a - b|.$$

The map d satisfies the axioms of a metric valued in T: for all $a, b, c \in R$,

1. Positive definiteness: $d(a, b) = 0_T$ if and only if $a = b$, by positive definiteness of $|\cdot|$.
2. Symmetry: $d(a, b) = |a-b| = |-1_R| \cdot |b-a| = |b-a| = d(b, a)$, since $|-1_R| = 1_T$ follows from multiplicativity ($|-1_R|^2 = |(-1_R)^2| = |1_R| = 1_T$) and positive definiteness.
3. Triangle inequality: $d(a, c) = |a-c| = |(a-b)+(b-c)| \leq |a-b|+|b-c| = d(a, b)+d(b, c)$, by subadditivity of $|\cdot|$.

In the case that T is not explicit, the absolute value $|\cdot|$ is almost always taken to be valued in $\mathbb{R}_{\geq 0}$, so that d takes values in $\mathbb{R}_{\geq 0}$.

Definition B.1.5. Let F be a field. An absolute value $|\cdot|$ on F (Definition B.1.1) is said to be *non-archimedean* if

$$|x + y| \leq \max(|x|, |y|) \quad \text{for all } x, y \in F$$

and is said to be *archimedean* otherwise.

Equivalently, $|\cdot$ is non-archimedean if and only if its induced metric (Definition B.1.4) is non-archimedean.

In particular, a *non-archimedean valued field* (resp. *archimedean valued field*) is a valued field (Definition B.1.1) whose valuation is non-archimedean (resp. archimedean)

Definition B.1.6 (Topology on a commutative ring induced by an absolute value)**.** Let T be an ordered semiring, and let R be a commutative ring (Definition C.1.1) equipped with an absolute value valued in T (Definition B.1.1):

$$|\cdot| : R \to T.$$

The *topology induced by $|\cdot|$ on R* is defined by declaring a subset $U \subseteq R$ to be open if for every $x \in U$, there exists $\varepsilon \in T_{>0}$ such that

$$B(x, \varepsilon) := \{y \in R : |y - x| < \varepsilon\} \subseteq U.$$

The set $B(x, \varepsilon)$ is called the *open ball of radius ε around x*.

In the case that $T = \mathbb{R}_{\geq 0}$, we recover the standard definition of the topology induced by an absolute value with real-valued radii.

Equivalently, the topology induced by $|\cdot|$ on R is the topology induced by (Definition C.2.3) the metric induced by $|\cdot|$ (Definition B.1.4).

If R is also a field, this topology makes $(R, \mathcal{T}_{|\cdot|})$ a topological field.

B.2. **Valuations on fields.**

Definition B.2.1 (Valuation on a field)**.** Let K be a field. A *valuation on K* is a function

$$v : K \to \Gamma \cup \{\infty\},$$

where $(\Gamma, +, \leq)$ is a totally ordered abelian group and ∞ is an element greater than all elements of Γ, satisfying for all $x, y \in K$:

1. $v(x) = \infty$ if and only if $x = 0$,
2. $v(xy) = v(x) + v(y)$,
3. $v(x + y) \geq \min\{v(x), v(y)\}$.

Alternatively (and essentially equivalently), a valuation on K is also defined as a function $v : K^\times \to \Gamma$ with properties (2) and (3) and is extended into a function $v : K \to \Gamma \cup \{\infty\}$ by setting $v(0) = \infty$.

The pair (K, v) is called a *valued field* (or less often a *valuation field*).

One usually calls the image $v(\Gamma)$ of v the *value group of v*. The *(rational) rank of v* is the dimension of the $\mathbb{Q}$-vector space $v(\Gamma) \otimes_{\mathbb{Z}} \mathbb{Q}$.

If Γ is embeddable into $(\mathbb{R}, +, \leq)$, then the valuation v induces (Definition B.2.4) a nonarchimedean absolute value (Definition B.1.5).

Definition B.2.2 (Valuation Ring of a Valued Field)**.** Let K be a field equipped with a valuation (Definition B.2.1) (♠ TODO: totally ordered abelian group) $v : K \to \Gamma$, where Γ is a totally ordered abelian group.

The *valuation ring* or *ring of integers of the valued field (K, v)* is the subring

$$\mathcal{O}_v := \{x \in K \mid v(x) \geq 0\},$$

where 0 is the neutral element of Γ. This ring $\mathcal{O}_v$ is a local ring (Definition C.1.5) with maximal ideal (Definition C.1.4)

$$\mathfrak{m}_v := \{x \in K \mid v(x) > 0\}.$$

Its *residue field* is defined as $\kappa_v := \mathcal{O}_v / \mathfrak{m}_v$.

Definition B.2.3 (Discrete valuation)**.** Let (K, v) be a valued field (Definition B.2.1). The valuation v is called *discrete* if the value group $\Gamma = v(K^\times)$ is isomorphic, as an ordered group, to $\mathbb{Z}$ (the integers with the usual order).

Definition B.2.4. Let K be a field and let $v : K \to \mathbb{R} \cup \{\infty\}$ be a valuation (Definition B.2.1) (of rank 1). Fix a real number c with $0 < c < 1$. The function

$$|x|_v := \begin{cases} c^{v(x)} & \text{if } x \in K^\times \\ 0 & \text{if } x = 0 \end{cases}$$

is a nonarchimedean absolute value (Definition B.1.1) on K.

Different values of c yield equivalent absolute values (Definition 2.2.4) and thus do not change the topology induced by $|\cdot|_v$ (Definition B.1.6)

Appendix C. Miscellaneous definitions

C.1. Abstract algebra.

Definition C.1.1. A *commutative (unital) ring* is a ring $(R, +, \cdot)$ such that $\cdot$ is a commutative operation, i.e. $a \cdot b = b \cdot a$.

For many writers (e.g. "commutative" algebraists or number theorists), a *ring* refers to a commutative ring as above.

Definition C.1.2. Let $(R, +, \cdot)$ be a not-necessarily commutative ring.

1. An element $a \in R$ is a *left zero-divisor* if there exists a nonzero $x \in R$ such that $ax = 0$. Otherwise, a is called *left regular* or *left cancellable*.
2. An element $a \in R$ is a *right zero-divisor* if there exists a nonzero $x \in R$ such that $xa = 0$. Otherwise, a is called *right regular* or *right cancellable*.
3. An element $a \in R$ is a *zero-divisor* if it is a left zero-divisor or a right zero-divsor.
4. An element $a \in R$ is a *two-sided zero-divisor* if it is both a left zero-divisor and a right zero-divsor.
5. An element $a \in R$ is *regular*, *cancellable*, or a *non-zero-divisor* if it is both left and right regular.

A zero-divisor of any kind that is not itself 0 is said to be a *nonzero zero divisor* or a *nontrivial zero divisor* of its kind.

Definition C.1.3. Let R be a ring and let $S \subseteq Z(R)$ be a multiplicative subset contained in the center of R. The *localization of R at S*, denoted by $S^{-1}R$, is the ring whose elements are equivalence classes of pairs $(r, s) \in R \times S$ under the equivalence relation

$$(r, s) \sim (r', s') \iff \exists u \in S \text{ such that } u(sr' - s'r) = 0.$$

Write $\frac{r}{s}$ for the equivalence class of (r, s). Addition and multiplication on representatives are defined by

$$\frac{r}{s} + \frac{r'}{s'} = \frac{rs' + r's}{ss'},$$
$$\frac{r}{s} \cdot \frac{r'}{s'} = \frac{rr'}{ss'}.$$

The map $r \mapsto \frac{r}{1}$ defines a ring homomorphism; therefore, $S^{-1}R$ is naturally an R-algebra.

Given an element $f \in Z(R)$, the localization of R at $S = \{f^n : n \geq 0\}$ is denoted by R_f.

In the case that R is a commutative ring and if P is a prime ideal of R (Definition C.1.4), then $R_P := S^{-1}R$ with $S = R \setminus P$ is called the *localization of R at P*. It is a local ring (Definition C.1.5) whose maximal ideal (Definition C.1.4) is given by

$$S^{-1}P = \{\frac{p}{s} \in R_P : p \in P\}.$$

Definition C.1.4. Let R be a (not necessarily commutative) ring. A proper two-sided ideal $P \trianglelefteq R$ is called a *prime ideal* if the following equivalent conditions holds:

1. If I, J are left ideals and $IJ \subset P$, then $I \subset P$ or $J \subset P$.
2. If I, J are right ideals and $IJ \subset P$, then $I \subset P$ or $J \subset P$.
3. If I, J are two-sided idaels and $IJ \subset P$, then $I \subset P$ or $J \subset P$.
4. If $x, y \in R$ with $xRy \subset \mathfrak{p}$, then $x \in \mathfrak{p}$ or $y \in \mathfrak{p}$.

A proper left/right/two-sided ideal $M \subsetneq R$ is called *maximal* if there exists no other left/right/two-sided ideal $J \trianglelefteq R$ such that $M \subsetneq J \subsetneq R$. Equivalently,

- a left/right ideal M of R is maximal if and only if the quotient module R/M is a simple left/right R-module.
- a two-sided ideal M of R is maximal if and only if the quotient ring R/M is a simple ring.

Definition C.1.5. Let R be a ring with unity, not necessarily commutative. The ring R is called a *local ring* if it has a unique maximal left ideal (Definition C.1.4). In this case, R also has a unique maximal right ideal, and these coincide with the Jacobson radical $J(R)$ of R.

The unique maximal left (and right) ideal of a local ring R may sometimes be denoted by $\mathfrak{m}_R$.

The following are each equivalent conditions for R to be a local ring:

1. $R/J(R)$ is a division ring
2. $R = J(R) \cup R^\times$

Definition C.1.6 (Principal ideal ring/domain (PID))**.** Let R be a commutative unital ring (Definition C.1.1). Then R is a *principal ideal ring (PIR)* if every ideal of R is principal. If R is additionally an , then R is said to be a *principal ideal domain (PID)*

Definition C.1.7 (Unique factorization domain (UFD) Factorial ring)**.** An integral domain R is called a *unique factorization domain (UFD)* or *factorial ring* if every nonzero nonunit element of R can be factored as a product of irreducible elements uniquely up to order and units.

Definition C.1.8. Let R be an integral domain. Let $n \geq 1$ be an integer. The *function field over R in n variables $x_1, \ldots, x_n$* is the field $R(x_1, \ldots, x_n)$ defined as the field of fractions of the polynomial ring $R[x_1, \ldots, x_n]$, which is an integral domain .

$R(x_1, \ldots, x_n)$ is also called the *field of rational numbers over R in n variables*. An element of $R(x_1, \ldots, x_n)$ is called a *rational function over R in n variables*.

C.2. **Absolute values and norms.**

Definition C.2.1 (Locally compact)**.** Let $(X, \mathcal{T})$ be a topological space. X is *locally compact* if for every $x \in X$, there exists an open set $U \in \mathcal{T}$ containing x and a compact set $K \subseteq X$ such that $x \in U \subseteq K$.

Definition C.2.2 (Extended metric induced by an extended norm on a module over a commutative ring with absolute value)**.** Let T be an ordered semiring, let R be a commutative ring (Definition C.1.1) equipped with an absolute value valued in T (Definition B.1.1):

$$|\cdot| : R \to T,$$

and let M be a module over R. Let T^∞ be the ordered semiring with adjoined infinity obtained from T, and let

$$\|\cdot\| : M \to T^\infty$$

be an extended norm on M valued in T^∞.

The *extended metric induced by the extended norm* is the function

$$\boxed{d : M \times M \to T^\infty, \quad (x, y) \mapsto \|x - y\|}$$

defined by

$$d(x, y) := \|x - y\|.$$

The map d satisfies the axioms of an extended metric valued in T^∞: for all $x, y, z \in M$,

1. Positive definiteness: $d(x, y) = 0_T$ if and only if $\|x - y\| = 0_T$ if and only if $x - y = 0_M$ if and only if $x = y$.
2. Symmetry: $d(x, y) = \|x - y\| = \|(-1_R)(y - x)\| = |-1_R| \cdot \|y - x\| = \|y - x\| = d(y, x)$, where $|-1_R| = 1_T$ follows from multiplicativity of the absolute value on the ring R.
3. Triangle inequality: $d(x, z) = \|x - z\| = \|(x - y) + (y - z)\| \leq \|x - y\| + \|y - z\| = d(x, y) + d(y, z)$, by the triangle inequality for $\|\cdot\|$ with addition in T^∞.

If the range of $\|\cdot\|$ is contained in T (so that $\|\cdot\|$ is a norm rather than an extended norm), then d takes values in T and is a metric.

In the case that $R = F$ is a field and $M = V$ is a vector space over F, we recover the standard definition of an extended metric on V induced by a norm valued in T^∞. In particular, when $T = \mathbb{R}_{\geq 0}$, we recover the standard definition of a metric/extended metric on V taking values in $\mathbb{R}_{\geq 0}$ or $[0, \infty]$.

Definition C.2.3 (Topology induced by an extended metric valued in an ordered monoid with adjoined infinity)**.** Let T be an ordered commutative monoid, let T^∞ be the ordered monoid with adjoined infinity obtained from T, and let (X, d) be an extended metric space valued in T^∞.

1. Given $\varepsilon \in T_{>0}$ and $x \in X$, the set

$$\boxed{B(x, \varepsilon) := \{y \in X : d(x, y) < \varepsilon\}}$$

is called the *open ball of radius ε centered at x*.

2. The *topology induced by d on X* is the topology defined by declaring a subset $U \subseteq X$ to be open if for every $x \in U$, there exists $\varepsilon \in T_{>0}$ such that $B(x, \varepsilon) \subseteq U$. In particular, $B(x, \varepsilon)$ is an open set for this topology for all $x \in X$ and $\varepsilon \in T_{>0}$.

Definition C.2.4 (Topology induced by a norm on a module over a commutative ring with absolute value)**.** Let T be an ordered semiring, let R be a commutative ring (Definition C.1.1) equipped with an absolute value valued in T (Definition B.1.1):

$$|\cdot| : R \to T,$$

and let M be a module over R. Let T^∞ be the ordered semiring with adjoined infinity obtained from T, and let $\|\cdot\|$ be an extended norm on M valued in T^∞. The *topology induced by the extended norm $\|\cdot\|$ on M* is defined by declaring a subset $U \subseteq M$ to be open if for every $x \in U$, there exists $\varepsilon \in T_{>0}$ such that

$$B(x, \varepsilon) := \{y \in M : \|y - x\| < \varepsilon\}$$

is contained in U. The set $B(x, \varepsilon)$ is called the *open ball of radius ε around x*.

In the case that $R = F$ is a field and $M = V$ is a vector space over F, we recover the standard definition of the topology induced by a norm on a vector space.

Equivalently, the topology on M induced by the extended norm $\|\cdot\|$ is the topology on M induced by (Definition C.2.3) the extended metric $d : M \times M \to T^\infty$ valued in T^∞ induced by $\|\cdot\|$ (Definition C.2.2).

Definition C.2.5 (Extended norm on a vector space over a field with absolute value)**.** Let T be an ordered semiring, let F be a field equipped with an absolute value valued in T (Definition B.1.1):

$$|\cdot| : F \to T,$$

and let V be a vector space over F. Let T^∞ be the ordered semiring with adjoined infinity obtained from T.

An *extended norm on V valued in T^∞* is a function

$$\|\cdot\| : V \to T^\infty$$

satisfying for all $x, y \in V$ and all scalars $\alpha \in F$:

1. Positive definiteness: $\|x\| = 0_T$ if and only if $x = 0$.
2. Homogeneity: $\|\alpha x\| = |\alpha| \cdot \|x\|$, where the multiplication on the right is in T^∞.
3. Triangle inequality: $\|x + y\| \leq \|x\| + \|y\|$, where addition and order are those of T^∞.

A vector space with an extended norm valued in T^∞ over a field with absolute value is called an *extended normed (vector) space*.

If the range of the extended norm is contained in T, then the extended norm is a *norm* in the usual sense and V may be called a *normed vector space* or a *normed space*.

In the case that $T = \mathbb{R}_{\geq 0}$, we recover the standard definition of an extended norm taking values in $\mathbb{R}_{\geq 0}$ or $[0, \infty]$.

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